PROJECT READOUT

Online Retail Customer Segmentation

A data-driven approach to retail growth and targeted marketing.

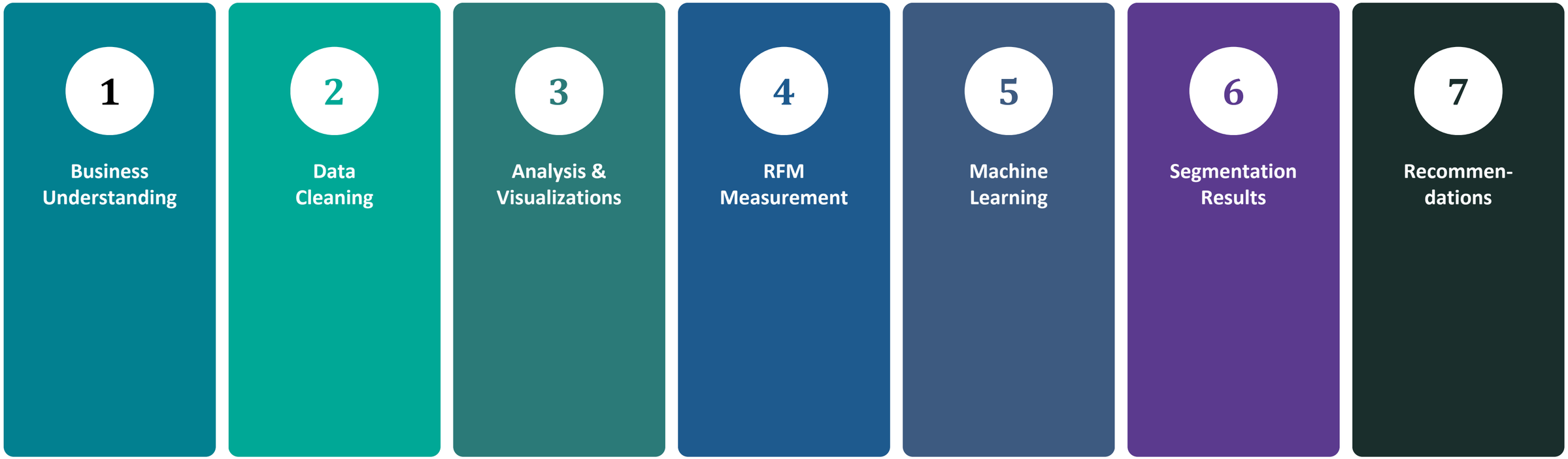
THE ANALYSTS

Meet the Team



- Esraa Yasser
- Aliaa Mohammed
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Our Approach, in 7 Phases



Each phase builds on the last: from a business question, to clean data, to a validated answer.

PHASES 1 & 2

Business & Data Understanding

Moving beyond generic transactions to build a trustworthy foundation.

PHASE 1: BUSINESS UNDERSTANDING

Beyond Transactions: The Core Challenge

- **The Problem:** Generic marketing ignores true customer value.
- **The Questions:** Which customers matter most? Who buys consistently? Who is at risk?
- **The Risk:** A generic £5 discount fails on a £5,000/year customer.
- **The Goal:** Actionable segments for targeted retention.



PHASE 2

Data Cleaning Pipeline

541k
Raw Transactions

389k
Clean Transactions

4.3k
Profiled Customers

Data recovered through order-matching. Guest purchases retained separately.

PHASE 2: DATA CLEANING

The Missing ID Dilemma

⚠ The Problem

155k

Missing Customer IDs

Approximately 28% of the raw data lacked identification. Dropping them meant losing massive insights.

The Solution

19k

Identities Recovered

Used shared Invoice Numbers to safely trace and map transactions back to known customers without guesswork.

PHASE 2: DATA CLEANING

Structuring for Success

The "Guest Customer" Strategy

- **Avoided deletion:** Isolated unidentifiable rows instead of dropping them.
- **Preserved insights:** Retained 16% of real revenue for general analysis.
- **Protected RFM:** Kept guests out of ML to prevent massive feature distortion.

1. Purchases

Cleaned and ready for RFM Modeling.

2. Returns

Isolated for cancellation analysis.

3. Guests

Preserved for global revenue tracking.

PHASE 3

Analysis & Visualizations

Looking at the shape of the business before building any models.

Online Retail — Business Snapshot

A quick view of the business before moving to customer segmentation

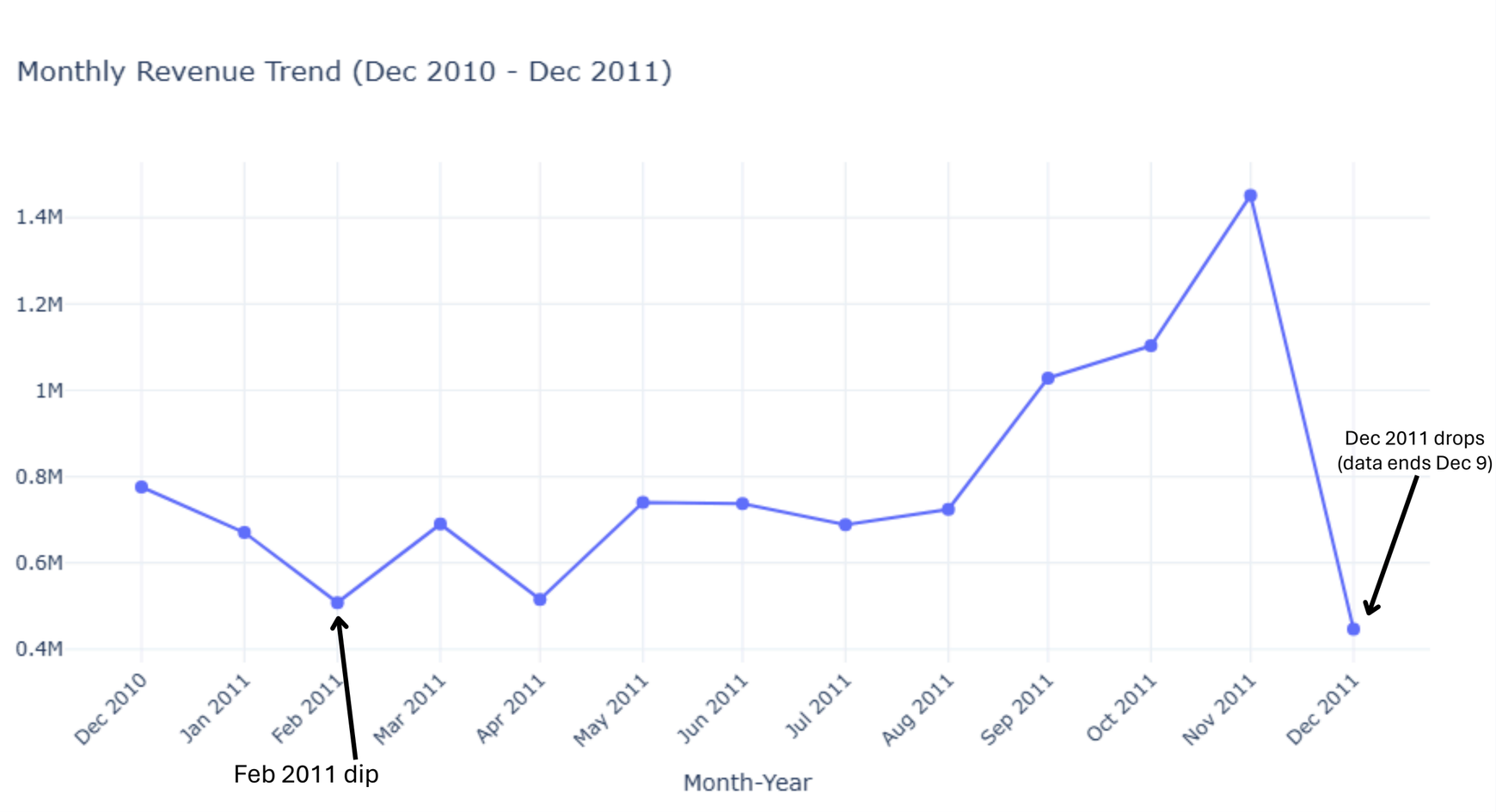
What does the business look like?

Total Revenue	Orders	CUSTOMERS	Average Order Value	Markets	Unique Products
£10.08M	19,702	4,332	£511.58	38	3,912

A UK-based online retailer with a broad product range and customers across 38 markets.

1. Revenue Trend — When Does the Business Peak?

Monthly revenue using all purchases (registered + guests)



Business readout

- Revenue rises gradually from March to November 2011.
- November peaks at ~£1.45M.
- February records the lowest monthly revenue at ~£508K.
- December is ~£0.45M, but the data ends on Dec 9, an incomplete month.
- The year-end revenue peak suggests an opportunity for strategic promotions.

Revenue strengthens toward the end of the year, with November showing the strongest monthly performance.

2. Product Performance — Value ≠ Volume

Top products by revenue, with quantity and unit-price context



What the chart tells us

REGENCY CAKESTAND 3 TIER:

- £165,987 revenue;
- 13,212 units;
- £13.95 unit price.

MEDIUM CERAMIC TOP STORAGE JAR:

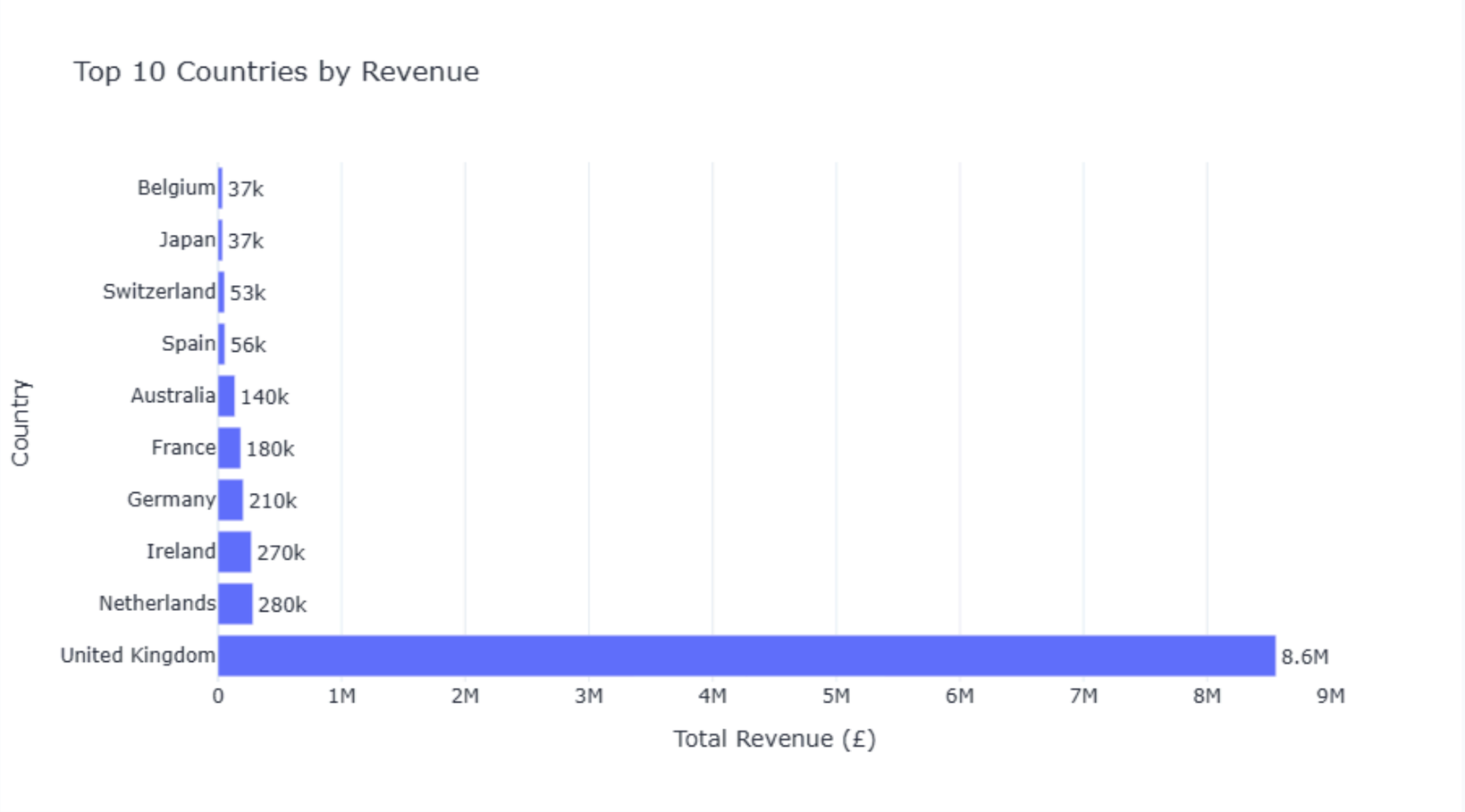
- £81,701 revenue
- 78,033 units
- £1.47 unit price.

Takeaway: high volume does not necessarily mean high revenue

High volume ≠ High revenue

3. Market Concentration — Where Does Revenue Come From?

Revenue by country highlights the geographic concentration of the business.



Key findings

- **UK: £8.56M—87.1% of revenue.**
- **Netherlands: £283.9K — 2.9%.**
- **Ireland: £270.9K — 2.8%.**
- **Germany: £205.4K — 2.1%.**
- **France: £184.5K — 1.9%.**

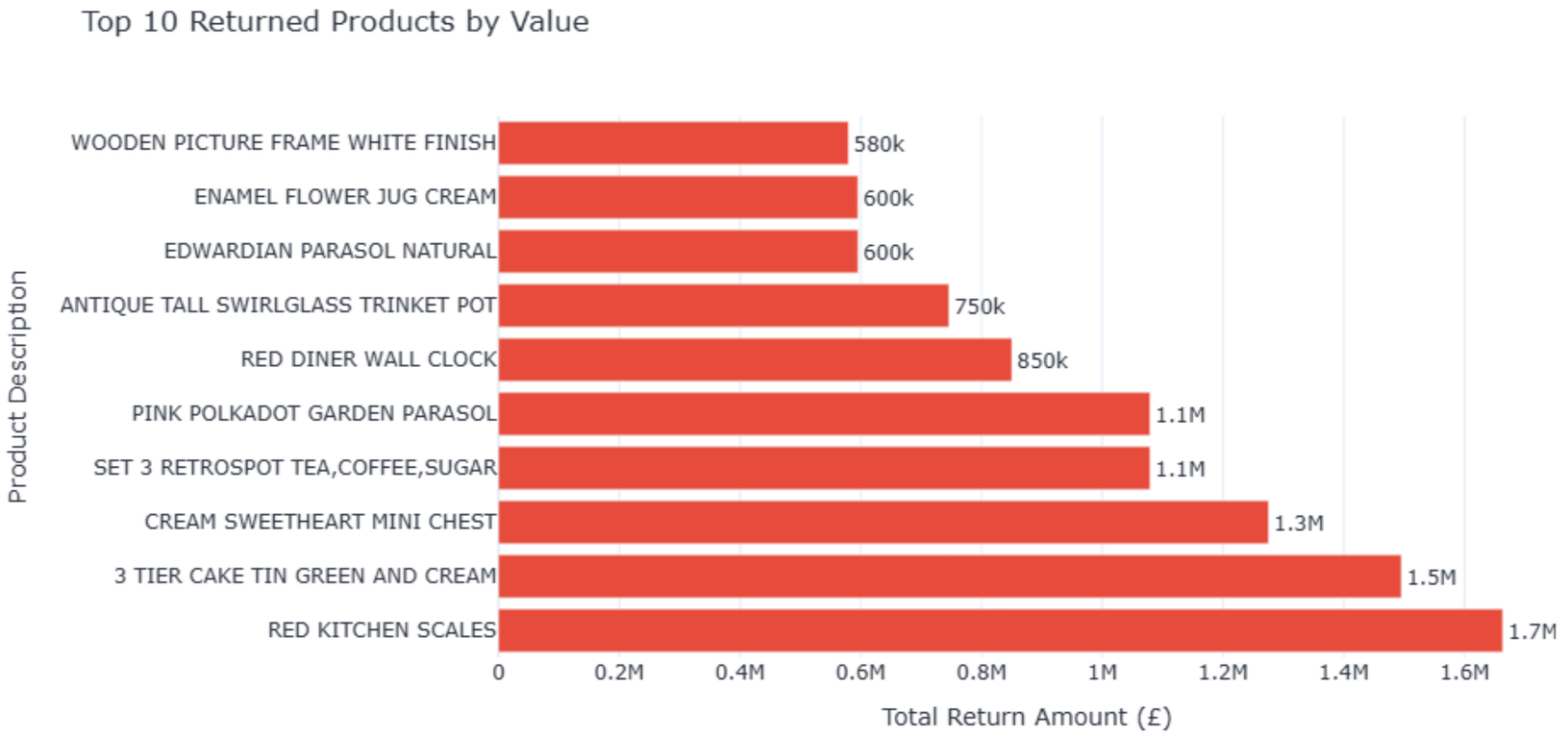
Top markets are mostly European; Australia and Japan also appear in the top 10.

Takeaway: the UK is the core market.

Business implication: the UK is the core market; international markets are smaller but identifiable for targeted growth.

4.EDA — Customer & Transaction Mix

What Makes Up the Transaction Data?



Returns

8,718 transaction (1.64%)

Guests (no account)

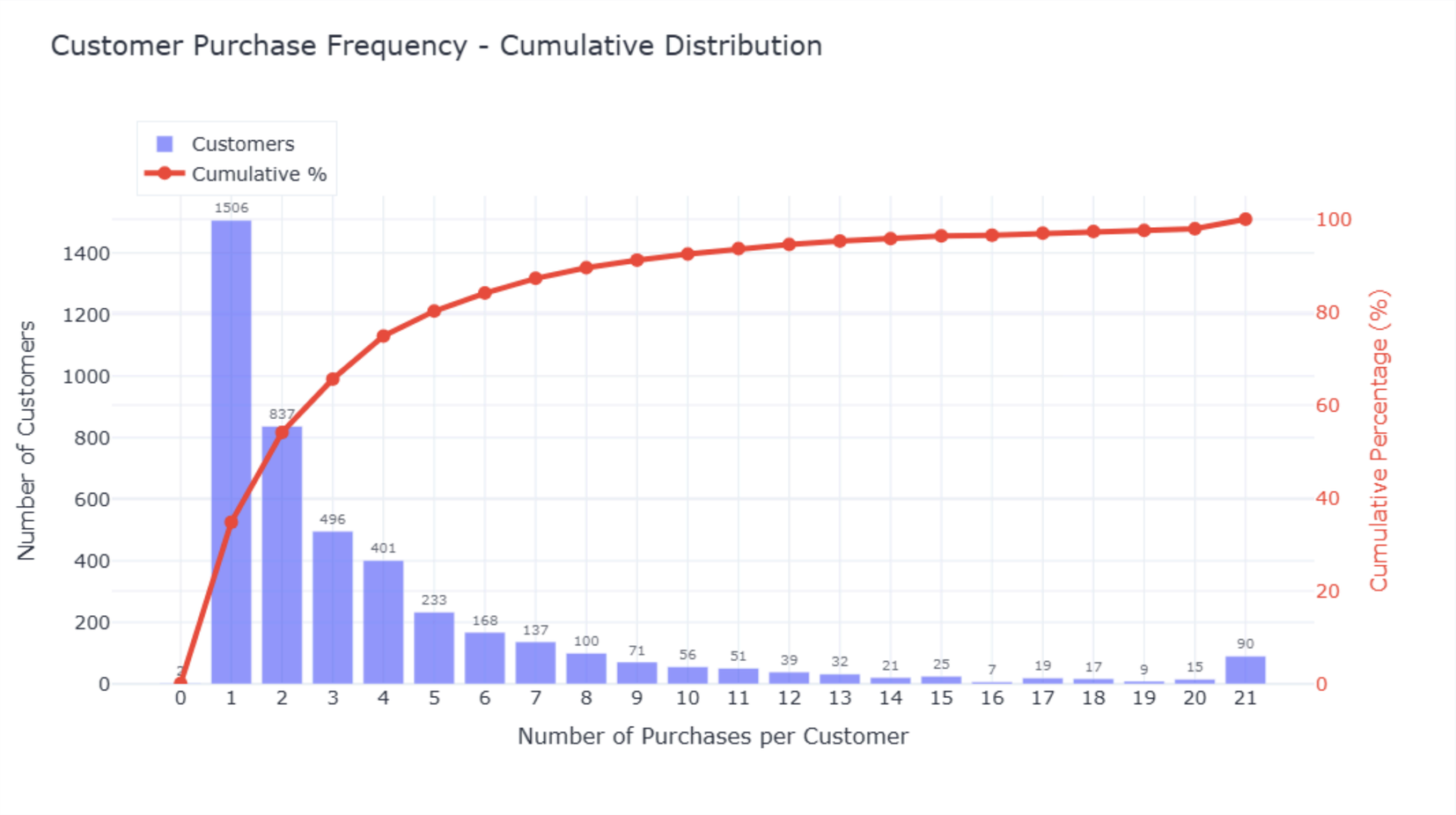
132,488 purchase transactions (34.1%)
£1.63M revenue (16.2%)

Registered customers

389,763 purchase transactions (65.9%)
£8.45M revenue (83.8%)

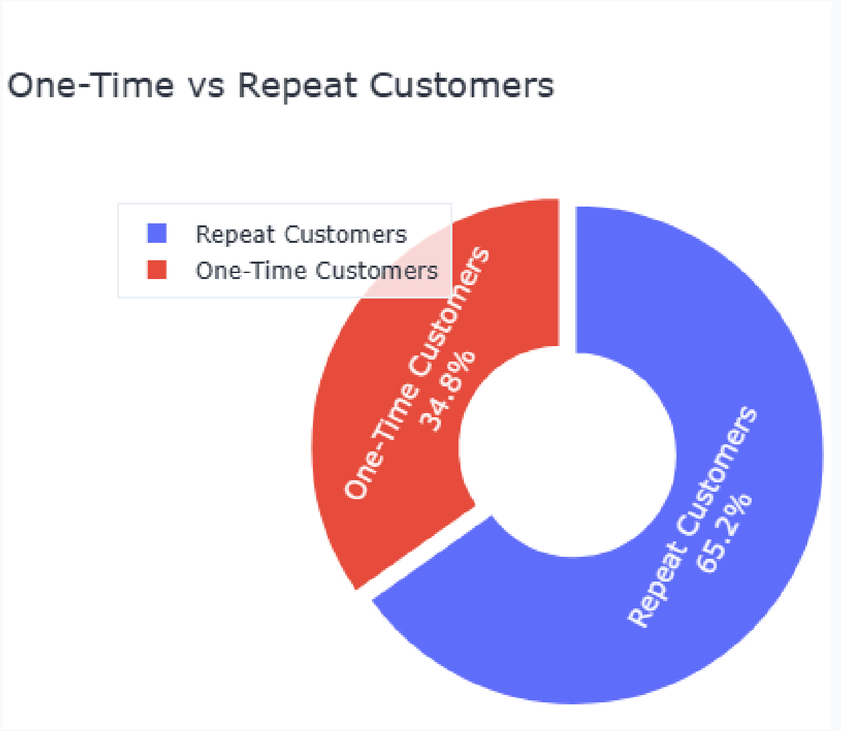
5. Customer Behavior — The Business Is Not One Customer Type

Purchase frequency is the point where EDA moves from transactions to customer behavior.



Customer-level findings

- **34.8% are one-time buyers.**
- **65.2% are repeat customers.**
- **Mean purchases/customer: 4.22; median: 2.**
- **Most frequent customer: 203 purchases.**
- **Top 10% of customers (433) account for 43.2% of all purchases.**
- **80.2% made only 1–5 purchases.**



EDA Takeaway — Why Customer Segmentation?

The EDA moves us from “what happened?” to “who behaves differently?”

Revenue

Nov 2011 ≈ £1.45M

Seasonality matters

Products

Value ≠ volume

Product strategy cannot be one-size-fits-all

Markets

UK = 87.1%

Revenue is geographically concentrated

Customers

65.2% repeat

Customer behavior is heterogeneous

Therefore, we need a customer-level representation that captures recency, frequency and monetary value.

TRANSACTIONS → CUSTOMER PROFILE (RFM) → SEGMENTS

PHASE 4

RFM Measurement

Turning thousands of transactions into three comparable numbers.

PHASE 4: RFM

Customer Profiles (RFM)

Recency

1 to 374 days

since last purchase.

Frequency

1 to 203

separate orders.

Monetary

£2.90 to £277k

total customer spend.

Finding: Wholesale outliers required mathematical log transformations to prevent model distortion.

What the Numbers Revealed

- **Most customers are small:** the typical customer bought once or twice and spent a modest amount — the majority cluster at the low end
- **A few customers are huge:** one customer alone spent over £277,000 across 70 separate orders — enough to pull the overall average far above what's typical
- **We checked it wasn't an error:** real, repeated, bulk purchases of low-cost items — a genuine wholesale-style buyer, not a data mistake
- **Why this matters:** we adjusted our math so this small handful of very large customers didn't distort the analysis for everyone else

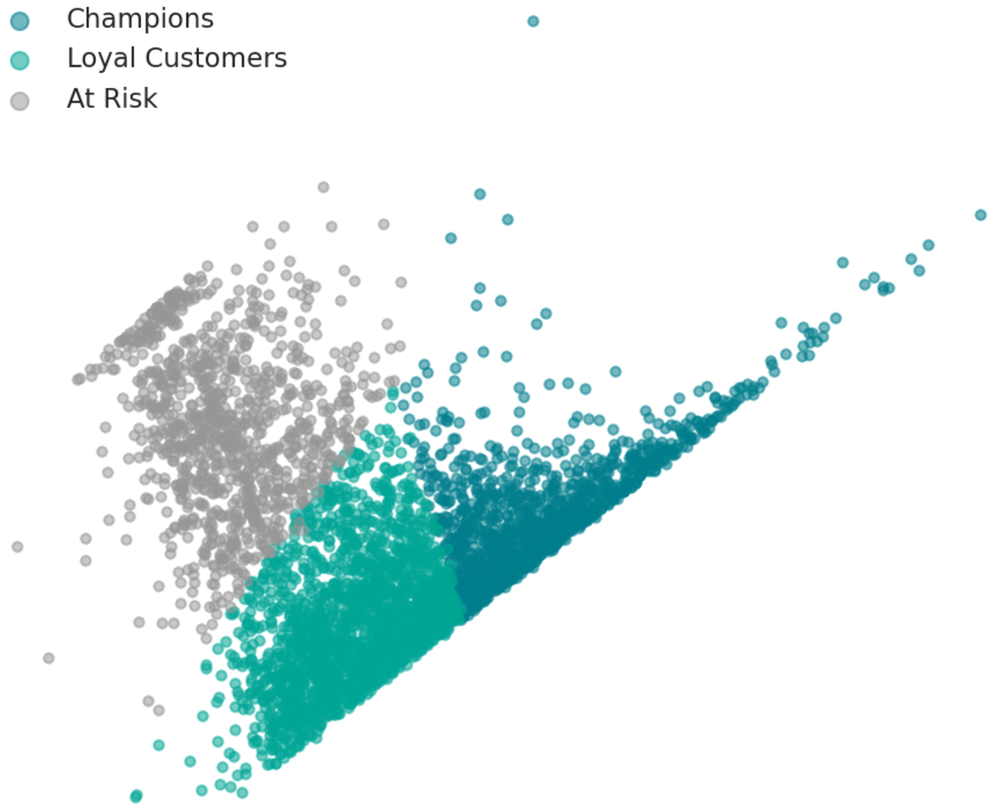
PHASES 5 & 6

Machine Learning & Segmentation

Letting algorithms find natural customer groupings we can trust.

How We Grouped Customers Automatically

- **The method:** a clustering algorithm looked at each customer's three RFM numbers and found natural groupings — without us telling it in advance who belongs where
- **Finding the right number of groups:** we tested splitting customers into anywhere from 2 to 10 groups, scoring each option on how well-separated the groups were
- **The result:** 3 groups gave the clearest, most distinct split



Simplified view of the 3 groups the algorithm found — each dot is a customer, colored by group.

Making Sure We Could Trust the Result

- **We didn't rely on one technique:** the same customers were regrouped using 3 other, completely different methods — and all of them pointed to the same underlying 3-group structure
- **An independent cross-check:** we separately calculated each customer's likely long-term value using a different formula entirely (not used in the grouping)
- **It matched:** 81% of our top segment landed in the 'highest value' bracket on this independent check, and 57% of our weakest segment landed in the 'lowest value' bracket

Four independent checks agreeing is strong evidence these groups reflect real differences — not a statistical coincidence.

PHASE 6

Three Key Segments

Champions

30.6% of base

Buy often. Spend big.

£5,145 Avg Spend

Loyal Customers

46.7% of base

Largest group. Modest spend.

£601 Avg Spend

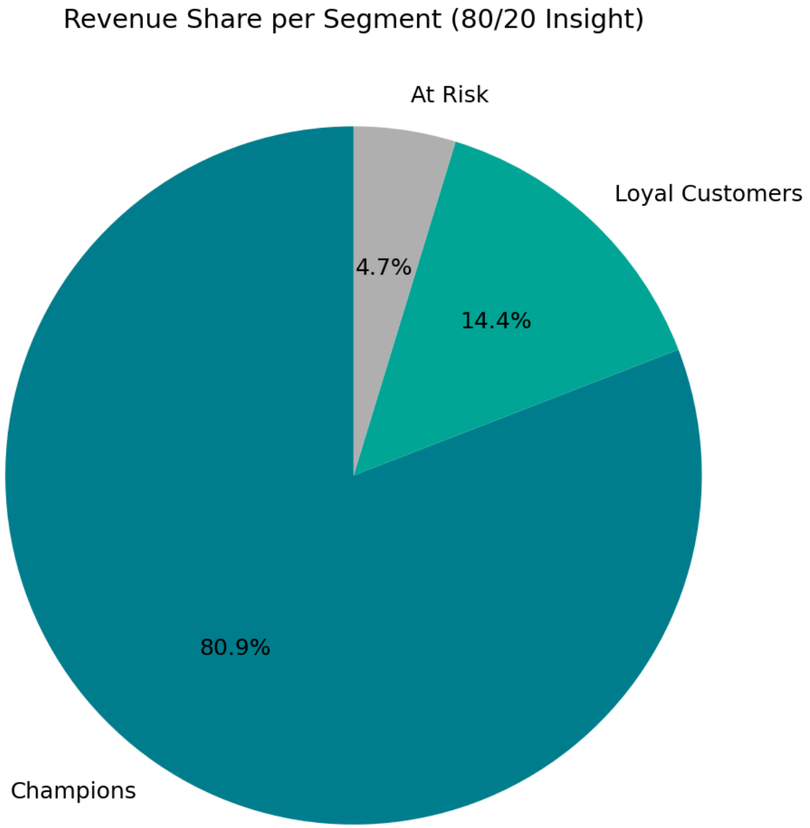
At Risk

22.7% of base

255 days inactive.

£405 Avg Spend

Where the Revenue Comes From



80.9%

of revenue from our registered, segmented customers comes from Champions — just 30.6% of that customer base

- A classic 80/20 pattern: a small group of customers funds most of the business
- Loyal Customers (the largest group) bring in only 14.4% — a clear growth opportunity
- At Risk customers contribute just 4.7% — little to lose by trying to win them back

PHASE 7

Recommendations

Turning validated segments into a concrete action plan.

PHASE 7: RECOMMENDATIONS

Strategic Next Steps



Protect Champions

Premium loyalty perks, early product access, and dedicated support.



Grow Loyal Customers

Targeted upsell campaigns and cross-selling to increase basket size.



Win Back At Risk

Personalized discount offers and decisive re-engagement emails.

Budget Principle: Allocate retention spend in proportion to segment value.

Thank You

Moving from generic marketing to data-driven growth.